



The maximum number of cliques in graphs with given fractional matching number and minimum degree

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ABSTRACT

Recently, Ma, Qian and Shi determined the maximum size of an n -vertex graph with given fractional matching number s and maximum degree at most d . Motivated by this result, we determine the maximum number of ℓ -cliques in a graph with given fractional matching number and minimum degree, which generalizes Shi and Ma's result about the maximum size of a graph with given fractional matching number and minimum degree at least one. We also determine the maximum number of complete bipartite graphs in a graph with prescribed fractional matching number and minimum degree.

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1. Introduction

We consider finite simple graphs and use standard terminology and notations [2]. Let G be a graph with vertex set $V(G)$ and edge set $E(G)$. We denote the cardinality of the vertex set by $n(G)$ and the cardinality of the edge set by $e(G)$. For a vertex v in a graph, we denote by $d(v)$ and $N(v)$ the degree of v and the neighborhood of v in G , respectively. For vertex disjoint graphs G and H , $G + H$ denotes the disjoint union of graphs G and H and $G \vee H$ denotes the join of G and H , which is obtained from the disjoint union $G + H$ by adding edges joining every vertex of G to every vertex of H . Denote by $\bar{G}$ the complement of a graph G . Let K_ℓ denote the complete graph of order ℓ and let K_{r_1, r_2} denote the complete bipartite graph with class sizes r_1, r_2 . We denote by $\delta(G)$ the minimum degree and denote by $\Delta(G)$ the maximum degree of a graph G . Let $\Gamma(v)$ denote the set of edges incident with v in G . Let $N(H, G)$ denote the number of copies of H in G ; e.g., $N(K_2, G) = e(G)$. Given a family of graphs $\mathcal{F}$, let $NM(H, \mathcal{F}) = \max \{N(H, F) \mid F \in \mathcal{F}\}$.

A matching is a set of pairwise nonadjacent edges of G . A fractional matching of a graph G is a function f assigning each edge with a real number in $[0, 1]$ so that $\sum_{e \in \Gamma(x)} f(e) \leq 1$ for each $x \in V(G)$. The fractional matching number of G , denoted by $\nu^*(G)$, is the maximum value of $\sum_{e \in E(G)} f(e)$ over all fractional matchings f . A matching is a special case of a fractional matching. It is known [12] that the fractional matching number is either an integer or a semi-integer, i.e., $2\nu^*(G)$ is an integer (see Theorem 2.1).

Since Turán proved his well-known theorem in 1941 [14], Turán-type problems have received a lot of attention [1,4–6,9,11]. In [5], Erdős and Gallai determined the maximum size of an n -vertex graph with matching number k .

Theorem 1.1 (Erdős and Gallai [5]). *Let n, k be two positive integers with $n \geq 2k + 1$. Let G be an n -vertex graph with matching number k . Then*

$$e(G) \leq \max \left\{ \binom{2k+1}{2}, \frac{k(2n-k-1)}{2} \right\}.$$

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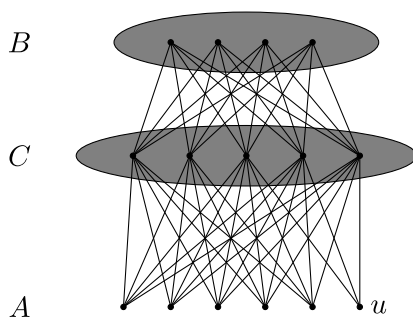


Fig. 1. $G(15, 7, 5)$ with $\delta = 3$.

It is natural to ask the same question by putting constraints on the graphs with given matching number. In [3], Chvátal and Hanson determined the maximum size of graphs with given matching number k and maximum degree at most d . By using the shifting method, Wang [15] determined the maximum number of copies of K_ℓ in an n -vertex graph with given matching number. In [4], Duan, Ning, Peng, Wang and Yang determined the maximum number of cliques in graphs with given minimum degree and matching number at most k . Recently, Liu and Zhang [7] determined the maximum number of copies of $K_{r_1, \dots, r_s}$ in graphs with given matching number and minimum degree at least k .

Along these results, Ma, Qian and Shi [10] determined the maximum size of an n -vertex graph with fractional matching number s and maximum degree at most d . As a corollary, they obtained the maximum size of a graph with a given number of vertices and fractional matching number.

Theorem 1.2 (Ma, Qian and Shi [10]). Let n , $2s$ and d be positive integers with $n > 2s$. Denote by $f(n, s, d) = \max\{e(G) : n(G) = n, v^*(G) = s, \Delta(G) \leq d\}$. If $2s$ is even, then

$$f(n, s, d) = \begin{cases} \max \left\{ \binom{2s}{2}, \left\lfloor \frac{s(n+d-s)}{2} \right\rfloor \right\} & \text{if } d \geq 2s - 1, n \leq d + s; \\ ds & \text{otherwise.} \end{cases}$$

If $2s$ is odd, then

$$f(n, s, d) = \begin{cases} \max \left\{ \binom{2s}{2}, \left\lfloor \frac{(s-\frac{3}{2})(n+d-s+\frac{3}{2})}{2} \right\rfloor + 3 \right\} & \text{if } d \geq 2s - 1, n \leq d + s - \frac{3}{2}; \\ \max \left\{ \binom{2s}{2}, d(s - \frac{3}{2}) + 3 \right\} & \text{if } d \geq 2s - 1, n \geq d + s - \frac{3}{2}; \\ \lfloor ds \rfloor & \text{if } d \leq 2s - 1. \end{cases}$$

It is natural to consider graphs with given fractional matching number and minimum degree. Recently, Shi and Ma [13] determined the maximum size of an n -vertex graph without isolated vertices for a given fractional matching number.

Notation 1.3. Let n , $2s$ and ℓ be positive integers with $n \geq 2s + 1 \geq 5$ and $\ell \geq 2$. Given a positive integer δ , for any integer t with $\delta \leq t \leq s$, denote by $\underline{G}(n, s, t)$ the graph obtained from $K_\ell \vee (K_{2s-2t} + \overline{K}_{n+t-2s})$ by deleting $t - \delta$ edges that are incident to one common vertex u in $\overline{K}_{n+t-2s}$, see Fig. 1 (the vertices in each gray ellipse induce a complete graph). Denote by $g_\ell(n, s, t)$ the number of copies of K_ℓ in $\underline{G}(n, s, t)$.

Let A be the set of vertices whose degree is at most t , C be the set of vertices whose degree is at least $n - 2$ in $\underline{G}(n, s, t)$ and let $B = V(\underline{G}(n, s, t)) \setminus (A \cup C)$. Note that the number of copies of K_ℓ in $B \cup C$ is $\binom{2s-t}{\ell}$, the number of copies of K_ℓ that contains a vertex in $A \setminus \{u\}$ and does not contain the vertex u is $\binom{t}{\ell-1}(n+t-2s-1)$ and the number of copies of K_ℓ that contains the vertex u is $\binom{\delta}{\ell-1}$. Therefore,

$$g_\ell(n, s, t) = \binom{2s-t}{\ell} + \binom{t}{\ell-1}(n+t-2s-1) + \binom{\delta}{\ell-1}. \quad (1)$$

Theorem 1.4 (Shi and Ma [13]). Let n , $2s$ be positive integers with $n \geq 2s + 1 \geq 5$. Let G be a graph of order n with fractional matching number s and minimum degree at least one.

If $2s$ is even, then

$$e(G) \leq \max \left\{ \binom{2s-2}{2} + n - 1, \binom{s}{2} + s(n-s) \right\}.$$

If $2s$ is odd, then

$$e(G) \leq \max \left\{ \binom{2s-2}{2} + n - 1, \binom{s-\frac{3}{2}}{2} + 3 + (s-\frac{3}{2})(n-s+\frac{3}{2}) \right\}.$$

Remark 1.5. The above equality holds if $\delta = 1$ and $G = G(n, s, 1)$ or $\delta = s$ and $G = G(n, s, s)$ when $2s$ is even, if $\delta = 1$ and $G = G(n, s, 1)$ or $\delta = s - \frac{3}{2}$ and $G = G(n, s, s - \frac{3}{2})$ when $2s$ is odd.

There are many graphs of order n with fractional matching number s and minimum degree δ . For example, the graph obtained from $K_{s, n-1-s} \cup K_1$ by adding δ edges between the vertex in K_1 and δ vertices in the partite set of size s is such a graph. Motivated by the above results, we determine the maximum number of copies of K_ℓ in n -vertex graphs with prescribed fractional matching number s and minimum degree δ .

Theorem 1.6. Let $n, 2s, \delta$ and ℓ be positive integers with $n \geq 2s + 1 \geq 5$ and $\ell \geq 2$. Let G be a graph of order n with fractional matching number s and minimum degree δ .

If $2s$ is even, then

$$N(K_\ell, G) \leq \max \{g_\ell(n, s, \delta), g_\ell(n, s, s)\}.$$

If $2s$ is odd, then

$$N(K_\ell, G) \leq \max \left\{ g_\ell(n, s, \delta), g_\ell \left(n, s, s - \frac{3}{2} \right) \right\}.$$

Theorem 1.6 is sharp as shown by the following remark.

Remark 1.7. Equality in **Theorem 1.6** holds if one of the following conditions holds:

- (1) If $G = G(n, s, \delta)$, $N(K_\ell, G) = g_\ell(n, s, \delta)$;
- (2) If $2s$ is even and $G = G(n, s, s)$, $N(K_\ell, G) = g_\ell(n, s, s)$;
- (3) If $2s$ is odd and $G = G(n, s, s - \frac{3}{2})$, $N(K_\ell, G) = g_\ell(n, s, s - \frac{3}{2})$.

Moreover, we find a lot of work on the maximum number of copies of K_{r_1, r_2} , see [8, 15–17]. In [15], Wang determined the maximum number of copies of K_{r_1, r_2} in bipartite graphs with a given matching number. In [16], Zhang determined the maximum number of copies of K_{r_1, r_2} in an n -vertex graph with given maximum size of linear forest and the minimum degree. Motivated by their work, we determine the maximum number of copies of K_{r_1, r_2} with prescribed fractional matching number and minimum degree.

Notation 1.8. Let $n, 2s, r_1$ and r_2 be positive integers with $n \geq 2s + 1 \geq 5$. Denote by $g_{r_1, r_2}(n, s, t)$ the number of copies of K_{r_1, r_2} in $G(n, s, t)$, where $G(n, s, t)$ is the graph defined in **Notation 1.3**.

Suppose that $r = r_1 + r_2$. Let $c = 1$ if $r_1 \neq r_2$, and $c = 2$ if $r_1 = r_2$. Let A be the set of vertices whose degree is at most t , C be the set of vertices whose degree is at least $n - 2$ in $G(n, s, t)$ and let $B = V(G(n, s, t)) \setminus (A \cup C)$. Note that the number of copies of K_{r_1, r_2} in $B \cup C$ is $\frac{1}{c} \binom{2s-t}{r} \binom{r}{r_1}$, the number of copies of K_{r_1, r_2} containing the vertex u is $\frac{1}{c} \sum_{j=1}^2 \binom{\delta}{r_j} \binom{n-r_j-1}{r-r_j-1}$ and for one partite set of K_{r_1, r_2} is in C , the other partite set contains a vertex in $A \setminus \{u\}$ and does not contain the vertex u , the number of copies of K_{r_1, r_2} is $\frac{1}{c} \sum_{j=1}^2 \binom{t}{r_j} \left[\binom{n-r_j-1}{r-r_j} - \binom{2s-t-r_j}{r-r_j} \right]$. Therefore,

$$\begin{aligned} g_{r_1, r_2}(n, s, t) &= \frac{1}{c} \sum_{j=1}^2 \binom{t}{r_j} \left[\binom{n-r_j-1}{r-r_j} - \binom{2s-t-r_j}{r-r_j} \right] + \frac{1}{c} \sum_{j=1}^2 \binom{\delta}{r_j} \binom{n-r_j-1}{r-r_j-1} \\ &\quad + \frac{1}{c} \binom{2s-t}{r} \binom{r}{r_1}. \end{aligned} \quad (2)$$

Theorem 1.9. Let $n, 2s, \delta, r_1$ and r_2 be positive integers with $n \geq 2s + 1 \geq 5$. Let G be a graph of order n with fractional matching number s and minimum degree δ .

If $2s$ is even, then

$$N(K_{r_1, r_2}, G) \leq \max \{g_{r_1, r_2}(n, s, \delta), g_{r_1, r_2}(n, s, s)\}.$$

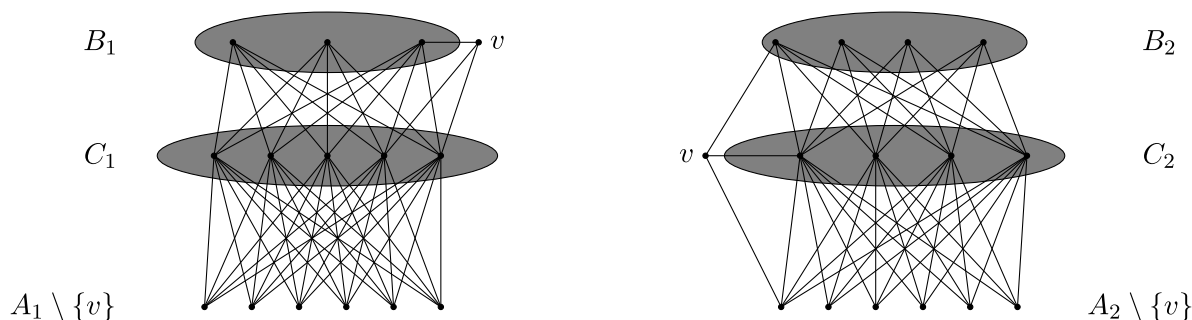


Fig. 2. $n = 15$, $s = 7$, $\delta = 3$, $G_1 \in \mathcal{F}_1(5)$ (left) and $G_2 \in \mathcal{F}_2(5)$ (right).

If $2s$ is odd, then

$$N(K_{r_1, r_2}, G) \leq \max \left\{ g_{r_1, r_2}(n, s, \delta), g_{r_1, r_2} \left(n, s, s - \frac{3}{2} \right) \right\}.$$

Theorem 1.9 is sharp as shown by the following remark.

Remark 1.10. Equality in Theorem 1.9 holds if one of the following conditions holds:

- (1) If $G = G(n, s, \delta)$, $N(K_{r_1, r_2}, G) = g_{r_1, r_2}(n, s, \delta)$;
- (2) If $2s$ is even and $G = G(n, s, s)$, $N(K_{r_1, r_2}, G) = g_{r_1, r_2}(n, s, s)$;
- (3) If $2s$ is odd and $G = G(n, s, s - \frac{3}{2})$, $N(K_{r_1, r_2}, G) = g_{r_1, r_2}(n, s, s - \frac{3}{2})$.

2. Proof of the results

To prove Theorem 1.6 and Theorem 1.9, we first need a well-known result, called fractional Tutte–Berge formula.

Theorem 2.1 ([12]). Let G be a graph of order n . Then

$$v^*(G) = \frac{1}{2} \left(n - \max_{T \subseteq V(G)} \{i(G - T) - |T|\} \right),$$

where $i(G - T)$ is the number of isolated vertices in $G - T$.

Let G be a graph of order n with fractional matching number s . By Theorem 2.1, there exists a subset T of $V(G)$ such that $s = \frac{1}{2}(n - (i(G - T) - |T|))$. Let $R \subseteq V(G)$ be the set of vertices that are not isolated in $G - T$. For convenience, we set $|T| = t$, $i(G - T) = q$ and $|R| = r$. Then $q = n - 2s + t$ and $r = n - t - q = 2s - 2t \neq 1$. Hence $t \neq s - \frac{1}{2}$. Since $t \leq n - q = n - (n - 2s + t) = 2s - t$, we have $t \leq s$. According to the previous analysis, we have the following lemma.

Lemma 2.2. Let G be a graph of order n with fractional matching number s . Then G is a subgraph of $K_t \vee (K_{2s-2t} + \overline{K_{n+t-2s}})$, with $t \leq s$ and $2s - 2t \neq 1$.

The following Pascal's Rule is useful throughout our proof. For any positive integers n, m with $n \geq m$, we have $\binom{n+1}{m} = \binom{n}{m} + \binom{n}{m-1}$.

Let $n, 2s$ be positive integers with $n \geq 2s + 1 \geq 5$. Given a positive integer δ , for any integer t with $\delta \leq t \leq s$, we denote

$$\mathcal{F}_1(t) = \{K_t \vee (K_{2s-2t} + \overline{K_{n+t-2s}}) - E_1 \mid E_1 \subseteq \Gamma(v), |E_1| = 2s - t - 1 - \delta, \text{ where } v \in V(K_{2s-2t})\}$$

and

$$\mathcal{F}_2(t) = \{K_t \vee (K_{2s-2t} + \overline{K_{n+t-2s}}) - E_2 \mid E_2 \subseteq \Gamma(v), |E_2| = n - 1 - \delta, \text{ where } v \in V(K_t)\}.$$

Two illustrative examples are presented in Fig. 2.

To prove Theorem 1.6, we need the following lemmas and proposition.

Lemma 2.3. Let $n, 2s$ and ℓ be positive integers with $n \geq 2s + 1 \geq 5$. Given a positive integer δ , we have

$$N(K_\ell, G(n, s, t)) \geq NM(K_\ell, \mathcal{F}_1(t)) \text{ for } \delta \leq t \leq s - \frac{1}{2}$$

and

$$N(K_\ell, G(n, s, t)) \geq NM(K_\ell, \mathcal{F}_2(t)) \text{ for } \delta \leq t \leq s.$$

Proof. Let $G_i(n, s, t)$ be a graph attaining the maximum number of copies of K_ℓ in $\mathcal{F}_i(t)$ for $i = 1, 2$.

First, we prove that $N(K_\ell, G(n, s, t)) \geq NM(K_\ell, \mathcal{F}_1(t))$ for $\delta \leq t \leq s - \frac{1}{2}$. Let A_1 be the set of vertices whose degree is at most t , C_1 be the set of vertices whose degree is at least $n - 2$ in $G_1(n, s, t)$ and let $B_1 = V(G_1(n, s, t)) \setminus (A_1 \cup C_1)$.

Note that the number of copies of K_ℓ in $B_1 \cup C_1$ is $\binom{2s-t-1}{\ell}$, the number of copies of K_ℓ that contains a vertex in $A_1 \setminus \{v\}$ is $\binom{t}{\ell-1}(n-2s+t)$ and the number of copies of K_ℓ that contains the vertex v is $\binom{\delta}{\ell-1}$. Thus,

$$N(K_\ell, G_1(n, s, t)) = \binom{2s-t-1}{\ell} + \binom{\delta}{\ell-1} + \binom{t}{\ell-1}(n-2s+t).$$

Since $t \leq s - \frac{1}{2}$, combining with Eq. (1), we have

$$N(K_\ell, G(n, s, t)) - N(K_\ell, G_1(n, s, t)) = \binom{2s-t-1}{\ell-1} - \binom{t}{\ell-1} \geq 0.$$

Next we prove that $N(K_\ell, G(n, s, t)) \geq NM(K_\ell, \mathcal{F}_2(t))$ for $\delta \leq t \leq s$. Let A_2 be the set of vertices whose degree is at most t , C_2 be the set of vertices whose degree is at least $n - 2$ in $G_2(n, s, t)$ and let $B_2 = V(G_2(n, s, t)) \setminus (A_2 \cup C_2)$.

Recall that v is the vertex with degree δ in $G_2(n, s, t)$. Then the number of copies of K_ℓ that contains the vertex v is at most $\binom{\delta}{\ell-1}$, the number of copies of K_ℓ that contains a vertex in $A_2 \setminus \{v\}$ and does not contain the vertex v is $\binom{t-1}{\ell-1}(n-2s+t)$, and the number of copies of K_ℓ in $B_2 \cup C_2$ is $\binom{2s-t-1}{\ell}$. Thus,

$$N(K_\ell, G_2(n, s, t)) \leq \binom{2s-t-1}{\ell} + \binom{\delta}{\ell-1} + \binom{t-1}{\ell-1}(n-2s+t).$$

Therefore, combining with Eq. (1), we have

$$\begin{aligned} & N(K_\ell, G(n, s, t)) - N(K_\ell, G_2(n, s, t)) \\ & \geq \binom{2s-t-1}{\ell-1} - \binom{t-1}{\ell-1} + \left(\binom{t}{\ell-1} - \binom{t-1}{\ell-1} \right) (n-2s+t-1) \\ & \geq 0, \end{aligned}$$

where the last inequality follows as $s \geq t$ and $n \geq 2s + 1$. $\square$

Lemma 2.4. Let ℓ and $2s$ be positive integers. For positive integer t with $t \leq 2s$, $\binom{2s-t}{\ell}$ is a convex function of t .

Lemma 2.5. Let n , $2s$ and ℓ be positive integers. For positive integer t , $f(t) = \binom{t}{\ell-1}(n+t-2s-1)$ is a convex function of t .

Proof. By direct calculation, we have

$$\begin{aligned} & f(t+1) + f(t-1) - 2f(t) \\ & = \binom{t+1}{\ell-1}(n+t-2s) + \binom{t-1}{\ell-1}(n+t-2s-2) - 2\binom{t}{\ell-1}(n+t-2s-1) \\ & = \left(\binom{t+1}{\ell-1} + \binom{t-1}{\ell-1} - 2\binom{t}{\ell-1} \right) (n+t-2s-1) + \binom{t+1}{\ell-1} - \binom{t-1}{\ell-1} \\ & = \left(\binom{t}{\ell-2} - \binom{t-1}{\ell-2} \right) (n+t-2s-1) + \binom{t+1}{\ell-1} - \binom{t-1}{\ell-1} \\ & \geq 0. \end{aligned}$$

This implies that $f(t)$ is a convex function of t . $\square$

By Lemmas 2.4 and 2.5, we have the following proposition.

Proposition 2.6. Let $n, 2s, \delta$ and ℓ be positive integers. For positive integer t with $t \leq s$,

$$g_\ell(n, s, t) = \binom{2s-t}{\ell} + \binom{t}{\ell-1} (n+t-2s-1) + \binom{\delta}{\ell-1}$$

is a convex function of t .

Proof of Theorem 1.6. Let G be a graph attaining the maximum number of copies of K_ℓ with fractional matching number s and minimum degree δ .

By Lemma 2.2, G is a subgraph of $K_t \vee (K_{2s-2t} + \overline{K_{n+t-2s}})$, with $t \leq s$ and $2s-2t \neq 1$. Since $\delta(G) = \delta$, it is clear that $\delta \leq t$, and hence G is a subgraph of $G(n, s, t)$, G_1 or G_2 , where $G_1 \in \mathcal{F}_1(t)$, $G_2 \in \mathcal{F}_2(t)$. Note that deleting any edge of a graph does not increase the number of copies of K_ℓ . So we may assume that $G = G(n, s, t)$, $G \in \mathcal{F}_1(t)$ or $G \in \mathcal{F}_2(t)$.

By Lemma 2.3, we have $N(K_\ell, G(n, s, t)) \geq NM(K_\ell, \mathcal{F}_1(t))$ for any t with $\delta \leq t \leq s - \frac{1}{2}$ and $N(K_\ell, G(n, s, t)) \geq NM(K_\ell, \mathcal{F}_2(t))$ for any t with $\delta \leq t \leq s$. In particular, if $G \in \mathcal{F}_1(s)$, then $\delta = s$ and hence $G = G(n, s, s)$. By the maximality of G , we may assume that $G = G(n, s, t)$ for some positive integers t with $\delta \leq t \leq s$.

Case 1. $2s$ is even. Now s is a positive integer. By Proposition 2.6, we have $t = s$ or $t = \delta$. If $t = s$, then $G = G(n, s, s)$ and hence $N(K_\ell, G) = g_\ell(n, s, s)$. If $t = \delta$, then $G = G(n, s, \delta)$ and hence $N(K_\ell, G) = g_\ell(n, s, \delta)$.

Case 2. $2s$ is odd. Since $t \neq s - \frac{1}{2}$, we have $\delta \leq t \leq s - \frac{3}{2}$. By Proposition 2.6, we have $t = \delta$ or $t = s - \frac{3}{2}$. If $t = s - \frac{3}{2}$, then $G = G(n, s, s - \frac{3}{2})$ and hence $N(K_\ell, G) = g_\ell(n, s, s - \frac{3}{2})$. If $t = \delta$, then $G = G(n, s, \delta)$ and hence $N(K_\ell, G) = g_\ell(n, s, \delta)$.

This completes the proof. $\square$

Next, to prove Theorem 1.9, we need the following lemmas and proposition.

Lemma 2.7. Let $n, 2s, r_1$ and r_2 be positive integers with $n \geq 2s + 1 \geq 5$. Given a positive integer δ ,

$$N(K_{r_1, r_2}, G(n, s, t)) \geq NM(K_{r_1, r_2}, \mathcal{F}_1(t)) \text{ for } \delta \leq t \leq s - \frac{1}{2}$$

and

$$N(K_{r_1, r_2}, G(n, s, t)) \geq NM(K_{r_1, r_2}, \mathcal{F}_2(t)) \text{ for } \delta \leq t \leq s.$$

Proof. Let $G_i(n, s, t)$ be a graph with the maximum number of copies of K_{r_1, r_2} in $\mathcal{F}_i(t)$ for $i = 1, 2$. Let $r = r_1 + r_2$. Consider the case $r_1 \neq r_2$.

First we prove that $N(K_{r_1, r_2}, G(n, s, t)) \geq NM(K_{r_1, r_2}, \mathcal{F}_1(t))$ for $\delta \leq t \leq s - \frac{1}{2}$. Let A_1 be the set of vertices whose degree is at most t , C_1 be the set of vertices whose degree is at least $n-2$ in $G_1(n, s, t)$ and let $B_1 = V(G_1(n, s, t)) \setminus (A_1 \cup C_1)$. We estimate the value of $N(K_{r_1, r_2}, G_1(n, s, t))$. Recall that v is the vertex with minimum degree δ in $G_1(n, s, t)$. Then the number of copies of K_{r_1, r_2} that contains the vertex v is at most $\sum_{j=1}^2 \binom{\delta}{r_j} \binom{n-r_j-1}{r-r_j-1}$. In $B_1 \cup C_1$, the number of copies of K_{r_1, r_2} is $\binom{2s-t-1}{r} \binom{r}{r_1}$. For one partite set of K_{r_1, r_2} is in C_1 , the other partite set contains a vertex in $A_1 \setminus \{v\}$ and does not contain the vertex v , the number of copies of K_{r_1, r_2} is $\sum_{j=1}^2 \binom{t}{r_j} \left[\binom{n-r_j-1}{r-r_j} - \binom{2s-t-r_j-1}{r-r_j} \right]$. Thus

$$\begin{aligned} N(K_{r_1, r_2}, G_1(n, s, t)) &\leq \sum_{j=1}^2 \binom{t}{r_j} \left[\binom{n-r_j-1}{r-r_j} - \binom{2s-t-r_j-1}{r-r_j} \right] \\ &\quad + \binom{2s-t-1}{r} \binom{r}{r_1} + \sum_{j=1}^2 \binom{\delta}{r_j} \binom{n-r_j-1}{r-r_j-1}. \end{aligned}$$

Combining with Eq. (2), we have

$$N(K_{r_1, r_2}, G(n, s, t)) - N(K_{r_1, r_2}, G_1(n, s, t)) \geq \binom{2s-t-1}{r-1} \binom{r}{r_1} - \sum_{j=1}^2 \binom{t}{r_j} \binom{2s-t-r_j-1}{r-r_j-1}.$$

Note that

$$\begin{aligned} \binom{2s-t-1}{r-1} \binom{r}{r_1} &\geq \sum_{j=1}^2 \binom{t}{r_j} \binom{2s-t-r_j-1}{r-r_j-1} \\ \iff \frac{(2s-t-1)! \cdot r!}{(r-1)! \cdot (2s-t-r)! \cdot (r-r_1)! \cdot r_1!} &\geq \sum_{j=1}^2 \frac{t! \cdot (2s-t-r_j-1)!}{r_j! \cdot (t-r_j)! \cdot (r-r_j-1)! \cdot (2s-t-r)!} \end{aligned}$$

$$\begin{aligned} \iff (2s-t-1)! \cdot r &\geq \frac{t! \cdot (2s-t-r_1-1)! \cdot r_2}{(t-r_1)!} + \frac{t! \cdot (2s-t-r_2-1)! \cdot r_1}{(t-r_2)!} \\ \iff 1 &\geq \frac{t(t-1) \cdots (t-r_1+1) \cdot r_2}{(2s-t-1)(2s-t-2) \cdots (2s-t-r_1) \cdot r} + \frac{t(t-1) \cdots (t-r_2+1) \cdot r_1}{(2s-t-1)(2s-t-2) \cdots (2s-t-r_2) \cdot r}. \end{aligned}$$

Since $t \leq s - \frac{1}{2}$, we have $2s-t-1 \geq t$, and hence

$$\frac{t(t-1) \cdots (t-r_j+1)}{(2s-t-1)(2s-t-2) \cdots (2s-t-r_j)} \leq 1 \text{ with } j = 1, 2.$$

Recall that $r_1 + r_2 = r$. Therefore, $N(K_{r_1, r_2}, G(n, s, t)) \geq NM(K_{r_1, r_2}, \mathcal{F}_1(t))$ for $\delta \leq t \leq s - \frac{1}{2}$.

Next we prove that $N(K_{r_1, r_2}, G(n, s, t)) \geq NM(K_{r_1, r_2}, \mathcal{F}_2(t))$ for $\delta \leq t \leq s - \frac{1}{2}$. Let A_2 be the set of vertices whose degree is at most t , C_2 be the set of vertices whose degree is at least $n-2$ in $G_2(n, s, t)$ and let $B_2 = V(G_2(n, s, t)) \setminus (A_2 \cup C_2)$. We estimate the value of $N(K_{r_1, r_2}, G_2(n, s, t))$. In $B_2 \cup C_2$, the number of copies of K_{r_1, r_2} is $\binom{2s-t-1}{r} \binom{r}{r_1}$. Recall that v is the vertex with minimum degree δ in $G_2(n, s, t)$. Then the number of copies of K_{r_1, r_2} that contains the vertex v is at most $\sum_{j=1}^2 \binom{\delta}{r_j} \binom{n-r_j-1}{r-r_j-1}$. For one partite set of K_{r_1, r_2} is in C_2 , the other partite set contains a vertex in $A_2 \setminus \{v\}$ and does not contain the vertex v , the number of copies of K_{r_1, r_2} is $\sum_{j=1}^2 \binom{t-1}{r_j} \left[\binom{n-r_j-1}{r-r_j} - \binom{2s-t-r_j-1}{r-r_j} \right]$. Thus,

$$\begin{aligned} N(K_{r_1, r_2}, G_2(n, s, t)) &\leq \sum_{j=1}^2 \binom{t-1}{r_j} \left[\binom{n-r_j-1}{r-r_j} - \binom{2s-t-r_j-1}{r-r_j} \right] \\ &\quad + \binom{2s-t-1}{r} \binom{r}{r_1} + \sum_{j=1}^2 \binom{\delta}{r_j} \binom{n-r_j-1}{r-r_j-1}. \end{aligned}$$

Therefore, combining with Eq. (2), we have

$$\begin{aligned} &N(K_{r_1, r_2}, G(n, s, t)) - N(K_{r_1, r_2}, G_2(n, s, t)) \\ &\geq \sum_{j=1}^2 \binom{t-1}{r_j-1} \binom{n-r_j-1}{r-r_j} - \sum_{j=1}^2 \binom{t}{r_j} \binom{2s-t-r_j}{r-r_j} + \sum_{j=1}^2 \binom{t-1}{r_j} \binom{2s-t-r_j-1}{r-r_j} \\ &\quad + \binom{2s-t-1}{r-1} \binom{r}{r_1} \\ &\geq \sum_{j=1}^2 \binom{t-1}{r_j-1} \binom{2s-t-r_j-1}{r-r_j} + \sum_{j=1}^2 \binom{t-1}{r_j} \binom{2s-t-r_j-1}{r-r_j} - \sum_{j=1}^2 \binom{t}{r_j} \binom{2s-t-r_j}{r-r_j} \\ &\quad + \binom{2s-t-1}{r-1} \binom{r}{r_1} \\ &\geq \binom{2s-t-1}{r-1} \binom{r}{r_1} - \sum_{j=1}^2 \binom{t}{r_j} \binom{2s-t-r_j-1}{r-r_j-1} \\ &\geq 0. \end{aligned}$$

The second inequality follows as $n > 2s-t$ and the last inequality holds by a similar discussion as above. Hence, $N(K_{r_1, r_2}, G(n, s, t)) \geq N(K_{r_1, r_2}, G_2(n, s, t))$ for $\delta \leq t \leq s - \frac{1}{2}$.

Suppose that $t = s$. In this case, $G_2(n, s, s) - \{v\}$ is a subgraph of $G(n, s, s) - \{u\}$. Hence, $N(K_{r_1, r_2}, G_2(n, s, s) - \{v\}) \leq N(K_{r_1, r_2}, G(n, s, s) - \{u\})$. Note that the number of copies of K_{r_1, r_2} containing the vertex v in $G_2(n, s, s)$ is at most $\sum_{j=1}^2 \binom{\delta}{r_j} \binom{n-r_j-1}{r-r_j-1}$, which is exactly the number of copies of K_{r_1, r_2} containing the vertex u in $G(n, s, s)$. Thus $N(K_{r_1, r_2}, G_2(n, s, s)) \leq N(K_{r_1, r_2}, G(n, s, s))$. Therefore, we have $N(K_{r_1, r_2}, G(n, s, t)) \geq NM(K_{r_1, r_2}, \mathcal{F}_2(t))$ for $\delta \leq t \leq s$.

For the case $r_1 = r_2$, by the same discussion and deleting repeated graphs, it is easy to verify that the number of copies of K_{r_1, r_2} is half of the above. This completes the proof. $\square$

Lemma 2.8. Let $n, 2s, r_1, r_2$ and r be positive integers with $n \geq 2s+1$ and $r = r_1 + r_2$. For positive integer t with $t \leq s$,

$$h(t) = \binom{2s-t}{r} \binom{r}{r_1} + \sum_{j=1}^2 \binom{t}{r_j} \left[\binom{n-r_j-1}{r-r_j} - \binom{2s-t-r_j}{r-r_j} \right]$$

is a convex function of t .

Proof. By direct calculation, we have

$$\begin{aligned} h(t+1) - h(t) = & - \binom{2s-t-1}{r-1} \binom{r}{r_1} + \sum_{j=1}^2 \binom{t}{r_j-1} \binom{n-r_j-1}{r-r_j} \\ & - \sum_{j=1}^2 \left[\binom{t+1}{r_j} \binom{2s-t-r_j-1}{r-r_j} - \binom{t}{r_j} \binom{2s-t-r_j}{r-r_j} \right]. \end{aligned}$$

Since $\binom{2s-t-r_j}{r-r_j} = \binom{2s-t-r_j-1}{r-r_j} + \binom{2s-t-r_j-1}{r-r_j-1}$, we have

$$\begin{aligned} h(t+1) - h(t) = & - \binom{2s-t-1}{r-1} \binom{r}{r_1} + \sum_{j=1}^2 \binom{t}{r_j-1} \binom{n-r_j-1}{r-r_j} \\ & + \sum_{j=1}^2 \left[\binom{t}{r_j} \binom{2s-t-r_j-1}{r-r_j-1} - \binom{t}{r_j-1} \binom{2s-t-r_j-1}{r-r_j} \right]. \end{aligned}$$

Similarly, we have

$$\begin{aligned} h(t-1) - h(t) = & \binom{2s-t}{r-1} \binom{r}{r_1} - \sum_{j=1}^2 \binom{t-1}{r_j-1} \binom{n-r_j-1}{r-r_j} \\ & - \sum_{j=1}^2 \left[\binom{t-1}{r_j} \binom{2s-t-r_j}{r-r_j-1} - \binom{t-1}{r_j-1} \binom{2s-t-r_j}{r-r_j} \right]. \end{aligned}$$

Therefore,

$$\begin{aligned} & h(t+1) + h(t-1) - 2h(t) \\ = & \sum_{j=1}^2 \binom{t-1}{r_j-2} \binom{n-r_j-1}{r-r_j} + \sum_{j=1}^2 \left[\binom{t}{r_j} \binom{2s-t-r_j-1}{r-r_j-1} - \binom{t-1}{r_j} \binom{2s-t-r_j}{r-r_j-1} \right] \\ & + \sum_{j=1}^2 \left[\binom{t-1}{r_j-1} \binom{2s-t-r_j}{r-r_j} - \binom{t}{r_j-1} \binom{2s-t-r_j-1}{r-r_j} \right] + \binom{2s-t-1}{r-2} \binom{r}{r_1}. \end{aligned}$$

Using the Pascal's Rule repeatedly, we have

$$\begin{aligned} & h(t+1) + h(t-1) - 2h(t) \\ = & \sum_{j=1}^2 \binom{t-1}{r_j-2} \binom{n-r_j-1}{r-r_j} + \sum_{j=1}^2 \left[\binom{t-1}{r_j-1} \binom{2s-t-r_j-1}{r-r_j-1} - \binom{t-1}{r_j} \binom{2s-t-r_j-1}{r-r_j-2} \right] \\ & + \sum_{j=1}^2 \left[\binom{t-1}{r_j-1} \binom{2s-t-r_j-1}{r-r_j-1} - \binom{t-1}{r_j-2} \binom{2s-t-r_j-1}{r-r_j} \right] + \binom{2s-t-1}{r-2} \binom{r}{r_1} \\ \geq & \sum_{j=1}^2 \binom{t-1}{r_j-2} \binom{n-r_j-1}{r-r_j} - \sum_{j=1}^2 \left[\binom{t-1}{r_j-2} \binom{2s-t-r_j-1}{r-r_j} + \binom{t-1}{r_j} \binom{2s-t-r_j-1}{r-r_j-2} \right] \\ & + \binom{2s-t-1}{r-2} \binom{r}{r_1}. \end{aligned}$$

Since $n - r_j - 1 \geq 2s - t - r_j - 1$, we have

$$h(t+1) + h(t-1) - 2h(t) \geq \binom{2s-t-1}{r-2} \binom{r}{r_1} - \sum_{j=1}^2 \binom{t-1}{r_j} \binom{2s-t-r_j-1}{r-r_j-2}.$$

To prove $h(t+1) + h(t-1) \geq 2h(t)$, we only need to prove

$$\binom{2s-t-1}{r-2} \binom{r}{r_1} \geq \sum_{j=1}^2 \binom{t-1}{r_j} \binom{2s-t-r_j-1}{r-r_j-2}$$

$$\begin{aligned} &\Longleftrightarrow \frac{(2s-t-1)! \cdot r!}{(r-2)! \cdot (2s-t-r+1)! \cdot (r-r_1)! \cdot r_1!} \geq \sum_{j=1}^2 \frac{(t-1)! \cdot (2s-t-r_j-1)!}{r_j! \cdot (t-1-r_j)! \cdot (r-r_j-2)! \cdot (2s-t-r+1)!} \\ &\Longleftrightarrow 1 \geq \sum_{j=1}^2 \frac{(t-1)(t-2) \cdots (t-r+r_j) \cdot r_j \cdot (r_j-1)}{(2s-t-1) \cdots (2s-t-r+r_j) \cdot r \cdot (r-1)}. \end{aligned}$$

Since $2s-t \geq t$ and $r = r_1 + r_2$, the last inequality holds. Therefore, $h(t+1) + h(t-1) \geq 2h(t)$, as desired. $\square$

By Lemma 2.8, we have the following proposition.

Proposition 2.9. Let $n, 2s, r, r_1$ and r_2 be positive integers with $n \geq 2s+1$ and $r = r_1 + r_2$. For positive integer t with $t \leq s$, let $g(t) = c \cdot g_{r_1, r_2}(n, s, t)$, where $c = 1$ if $r_1 \neq r_2$, and $c = 2$ if $r_1 = r_2$. Then

$$g(t) = \binom{2s-t}{r} \binom{r}{r_1} + \sum_{j=1}^2 \binom{t}{r_j} \left[\binom{n-r_j-1}{r-r_j} - \binom{2s-t-r_j}{r-r_j} \right] + \sum_{j=1}^2 \binom{\delta}{r_j} \binom{n-r_j-1}{r-r_j-1}$$

is a convex function of t .

Proof of Theorem 1.9. Let G be a graph attaining the maximum number of copies of K_{r_1, r_2} with fractional matching number s and minimum degree δ . We first consider the case $r_1 \neq r_2$.

By Lemma 2.2, G is a subgraph of $K_t \vee (K_{2s-2t} + K_{n+t-2s})$, with $t \leq s$ and $2s-2t \neq 1$. Since $\delta(G) = \delta$, it is clear that $\delta \leq t$, and hence G is a subgraph of $G(n, s, t)$, G_1 or G_2 , where $G_1 \in \mathcal{F}_1(t)$, $G_2 \in \mathcal{F}_2(t)$. Note that deleting any edge of a graph does not increase the number of copies of K_{r_1, r_2} . So we may assume that $G = G(n, s, t)$, $G \in \mathcal{F}_1(t)$ or $G \in \mathcal{F}_2(t)$.

By Lemma 2.7, we have $N(K_{r_1, r_2}, G(n, s, t)) \geq NM(K_{r_1, r_2}, \mathcal{F}_1(t))$ for any t with $\delta \leq t \leq s - \frac{1}{2}$ and $N(K_{r_1, r_2}, G(n, s, t)) \geq NM(K_{r_1, r_2}, \mathcal{F}_2(t))$ for any t with $\delta \leq t \leq s$. In particular, if $G \in \mathcal{F}_1(s)$, then $\delta = s$ and hence $G = G(n, s, s)$. By the maximality of G , we may assume that $G = G(n, s, t)$ for some positive integers t with $\delta \leq t \leq s$.

Case 1. $2s$ is even. In this case, s is an integer. By Proposition 2.9, we have $t = \delta$ or $t = s$. If $t = \delta$, then $G = G(n, s, \delta)$ and hence $N(K_{r_1, r_2}, G) = g_{r_1, r_2}(n, s, \delta)$. If $t = s$, then $G = G(n, s, s)$ and hence $N(K_{r_1, r_2}, G) = g_{r_1, r_2}(n, s, s)$.

Case 2. $2s$ is odd. Since $t \neq s - \frac{1}{2}$, we have $\delta \leq t \leq s - \frac{3}{2}$. By Proposition 2.9, we have $t = \delta$ or $t = s - \frac{3}{2}$. If $t = \delta$, then $G = G(n, s, \delta)$, and hence $N(K_{r_1, r_2}, G) = g_{r_1, r_2}(n, s, \delta)$. If $t = s - \frac{3}{2}$, then $G = G(n, s, s - \frac{3}{2})$ and hence $N(K_{r_1, r_2}, G) = g_{r_1, r_2}(n, s, s - \frac{3}{2})$.

For the case $r_1 = r_2$, we have $c = 2$ in Eq. (2), by the same discussion, it is easy to verify that the number of copies of K_{r_1, r_2} is as desired. This completes the proof. $\square$

3. Conclusion

In this paper, we have determined the maximum number of copies of K_ℓ in an n -vertex graph with prescribed fractional matching number and minimum degree. Our result yields Shi and Ma's work in [13] about the maximum size of graphs with given fractional matching number and minimum degree at least one. Moreover, we have used a similar method to determine the maximum number of copies of K_{r_1, r_2} with prescribed fractional matching number and minimum degree. In future research, we believe that investigating the maximum number of ℓ -cliques in graphs with given fractional matching numbers and other parameters is also an interesting topic.

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Data availability

No data was used for the research described in the article.

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